

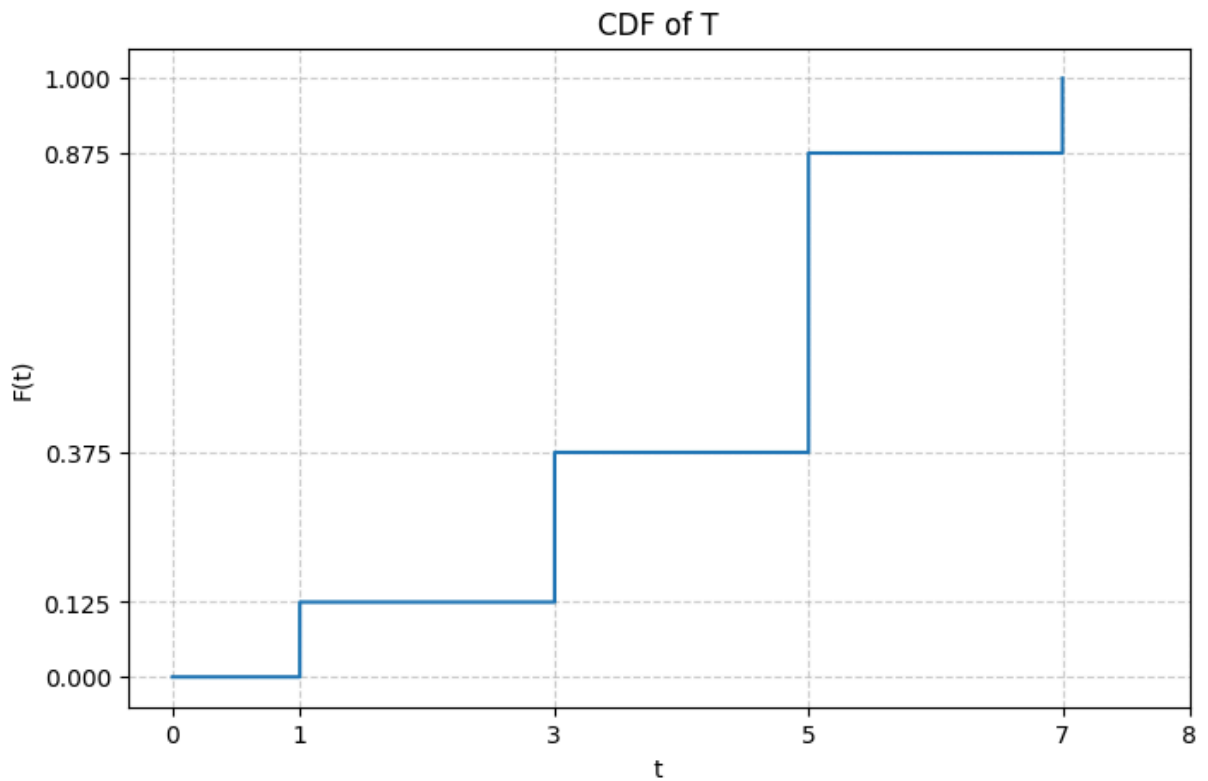
1.

(a)

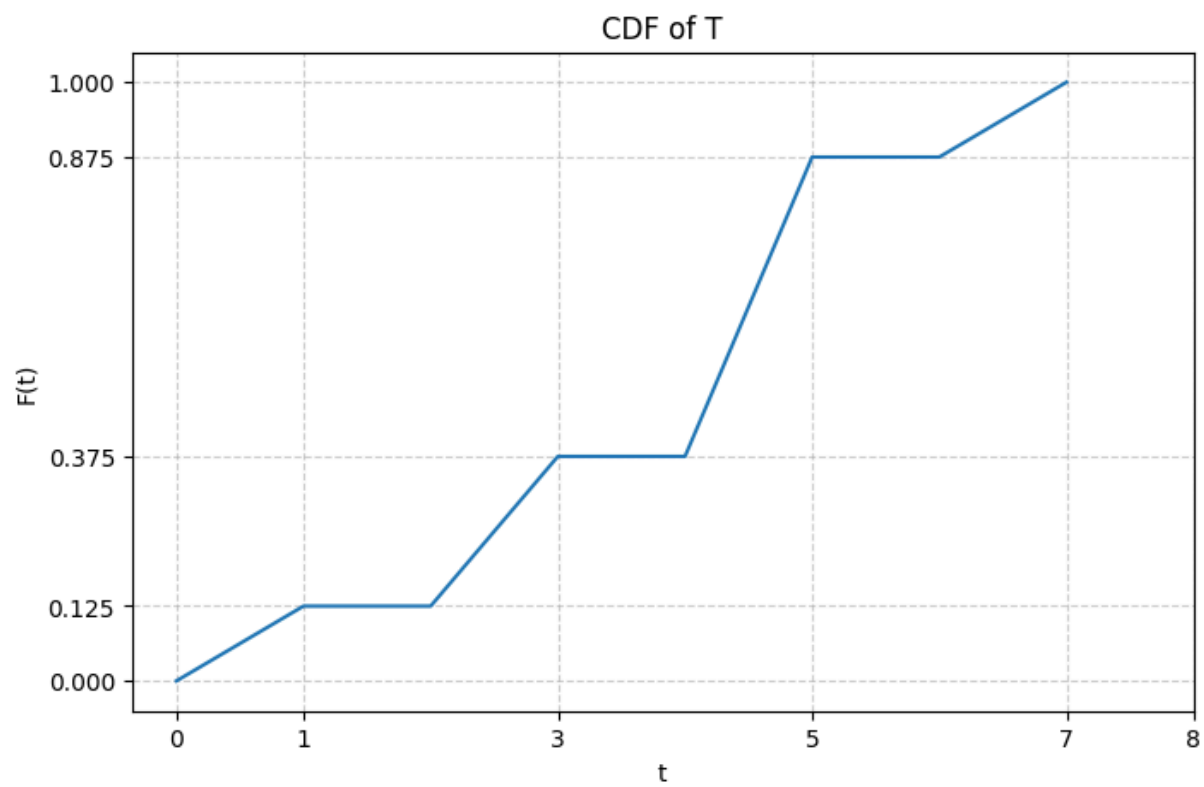
```
In [9]: import numpy as np
import matplotlib.pyplot as plt

def cumulative_distribution_function(t: float) -> float:
    if t < 1:
        return 0
    elif 1 <= t < 3:
        return 1/8
    elif 3 <= t < 5:
        return 3/8
    elif 5 <= t < 7:
        return 7/8
    else:
        return 1

t_values = np.linspace(0, 7, 1000)
F_values = [cumulative_distribution_function(t) for t in t_values]
plt.figure(figsize=(8, 5))
plt.step(t_values, F_values, where='post')
plt.xlabel('t')
plt.ylabel('F(t)')
plt.title('CDF of T')
plt.xticks([0, 1, 3, 5, 7, 8])
plt.yticks([0, 1/8, 3/8, 7/8, 1])
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```



```
In [8]: t_values = np.linspace(0, 7, 8)
F_values = [cumulative_distribution_function(t) for t in t_values]
plt.figure(figsize=(8, 5))
plt.plot(t_values, F_values)
plt.xlabel('t')
plt.ylabel('F(t)')
plt.title('CDF of T')
plt.xticks([0, 1, 3, 5, 7, 8])
plt.yticks([0, 1/8, 3/8, 7/8, 1])
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```



(b)

$$F(t) = \begin{cases} 0, & t < 1 \\ \frac{1}{8}, & 1 \leq t < 3 \\ \frac{3}{8}, & 3 \leq t < 5 \\ \frac{7}{8}, & 5 \leq t < 7 \\ 1, & t \geq 7 \end{cases}$$

$$P(T = t) = F(t) - \lim_{x \rightarrow t^-} F(x)$$

- **At $T = 1$:**

$$P(T = 1) = F(1) - F(0) = \frac{1}{8} - 0 = \frac{1}{8}$$

- **At $T = 3$:**

$$P(T = 3) = F(3) - F(2) = \frac{3}{8} - \frac{1}{8} = \frac{2}{8} = \frac{1}{4}$$

- **At $T = 5$:**

$$P(T = 5) = F(5) - F(4) = \frac{7}{8} - \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

- **At $T = 7$:**

$$P(T = 7) = F(7) - F(6) = 1 - \frac{7}{8} = \frac{1}{8}$$

T	P(T)
1	1/8
3	1/4
5	1/2
7	1/8

(c)

$$P(1.4 < T < 6) = F(6) - F(1.4)$$

$$F(6) \text{ falls in the range } 5 \leq t < 7, \text{ where } F(5) = \frac{7}{8}.$$

$$F(6) = F(5) = \frac{7}{8}$$

$$F(1.4) \text{ falls in the range } 1 \leq t < 3, \text{ where } F(1) = \frac{1}{8}.$$

$$F(1.4) = F(1) = \frac{1}{8}$$

$$P(1.4 < T < 6) = F(6) - F(1.4)$$

$$= \frac{7}{8} - \frac{1}{8} = \frac{6}{8} = \frac{3}{4}$$

$$P(1.4 < T < 6) = \frac{3}{4}$$

(d)

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- $A = \{T \leq 5\}$
- $B = \{T \geq 2\}$

$$P(T \leq 5 | T \geq 2) = \frac{P(2 \leq T \leq 5)}{P(T \geq 2)}$$

$$P(T \geq 2) = 1 - P(T < 2)$$

$$P(T < 2) = F(2) = F(1) = \frac{1}{8}$$

$$P(T \geq 2) = 1 - \frac{1}{8} = \frac{7}{8}$$

$$P(2 \leq T \leq 5) = P(T = 3) + P(T = 5)$$

$$P(T = 3) = \frac{1}{4}, \quad P(T = 5) = \frac{1}{2}$$

$$P(2 \leq T \leq 5) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$P(T \leq 5 | T \geq 2) = \frac{P(2 \leq T \leq 5)}{P(T \geq 2)}$$

$$= \frac{\frac{3}{4}}{\frac{7}{8}}$$

$$= \frac{3}{4} \times \frac{8}{7} = \frac{24}{28} = \frac{6}{7}$$

$$P(T \leq 5 | T \geq 2) = \frac{6}{7}$$

2. The data “student-por.csv” consists of Portuguese language course students final grades (column “G3”) in two schools “GP” (Gabriel Pereira) and “MS” (Mousinho da Silveira). Let X and Y stand for the grade “G3” in the two schools respectively. Note that X and Y are discrete variables taking possible values 0, 1, 2, . . . , 20.

```
In [7]: import pandas as pd
from pandas import read_csv
import numpy as np
import matplotlib.pyplot as plt
```

```

def compute_distribution(df, school):
    grades = df[df['school'] == school]['G3']
    distribution = grades.value_counts(normalize=True).sort_index()
    return distribution

def plot_distributions(dist_x, dist_y, title):
    plt.figure(figsize=(10, 5))
    if dist_x is not None:
        plt.bar(dist_x.index, dist_x.values, alpha=0.6, label='GP', color='b')
    if dist_y is not None:
        plt.bar(dist_y.index, dist_y.values, alpha=0.6, label='MS', color='r')
    plt.xlabel('Final Grade')
    plt.ylabel('Probability')
    plt.title(title)
    plt.legend()
    plt.show()

def compute_statistics(dist):
    values = np.array(dist.index)
    probabilities = np.array(dist.values)
    mean = np.sum(values * probabilities)
    variance = np.sum((values - mean) ** 2 * probabilities)
    std_dev = np.sqrt(variance)
    return mean, std_dev

def main():
    df = read_csv('student-por.csv' )

    dist_x = compute_distribution(df, 'GP')
    dist_y = compute_distribution(df, 'MS')

    print("(a)")
    plot_distributions(None, dist_y, 'Grade Distributions of GP')
    plot_distributions(dist_x, None, 'Grade Distributions of MS')
    plot_distributions(dist_x, dist_y, 'Grade Distributions of GP and MS')

    mean_x, std_x = compute_statistics(dist_x)
    mean_y, std_y = compute_statistics(dist_y)

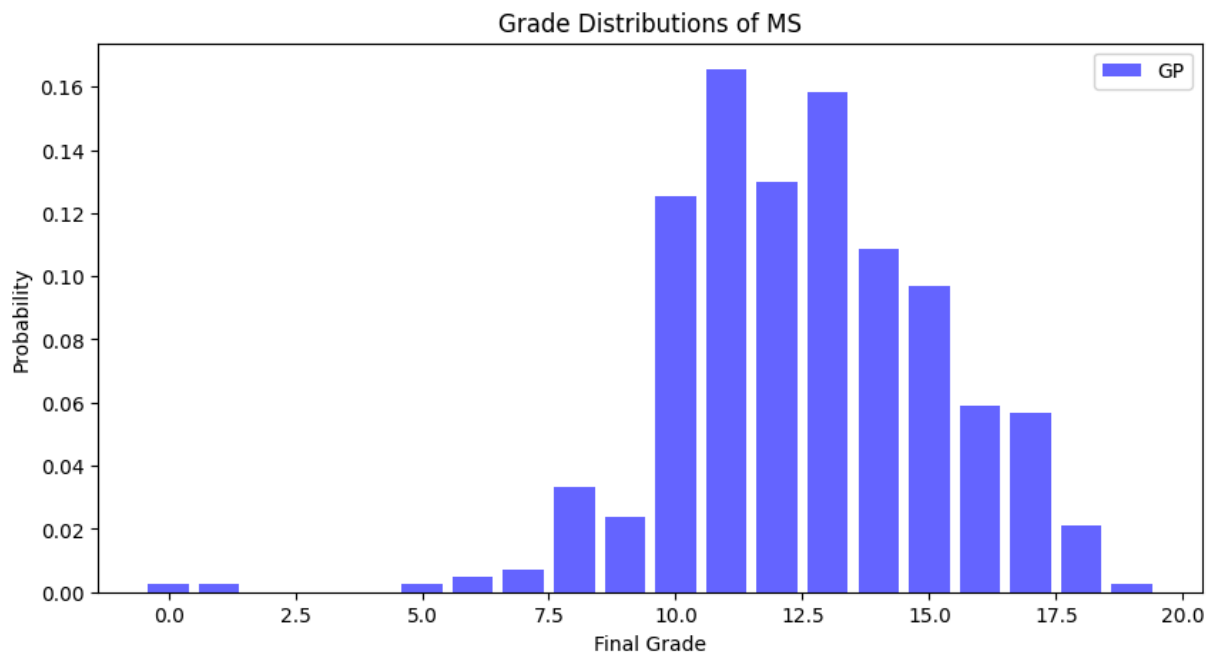
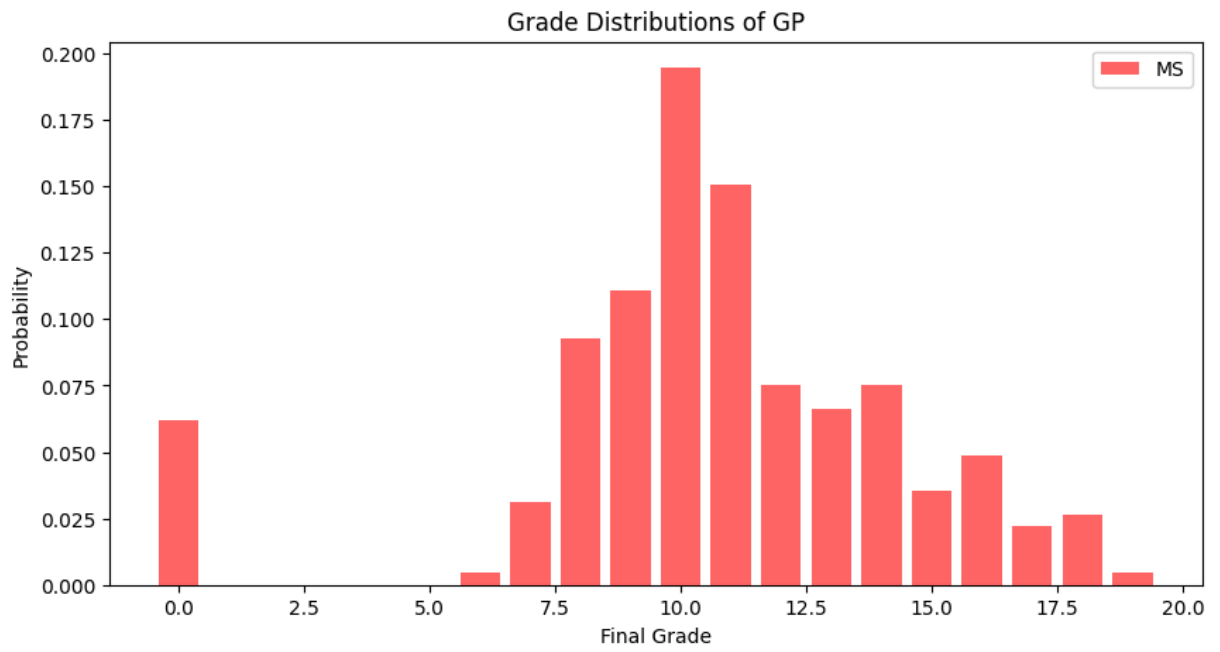
    print("(b)")
    print(f'GP: Mean = {mean_x:.2f}, Std Dev = {std_x:.2f}')
    print(f'MS: Mean = {mean_y:.2f}, Std Dev = {std_y:.2f}')

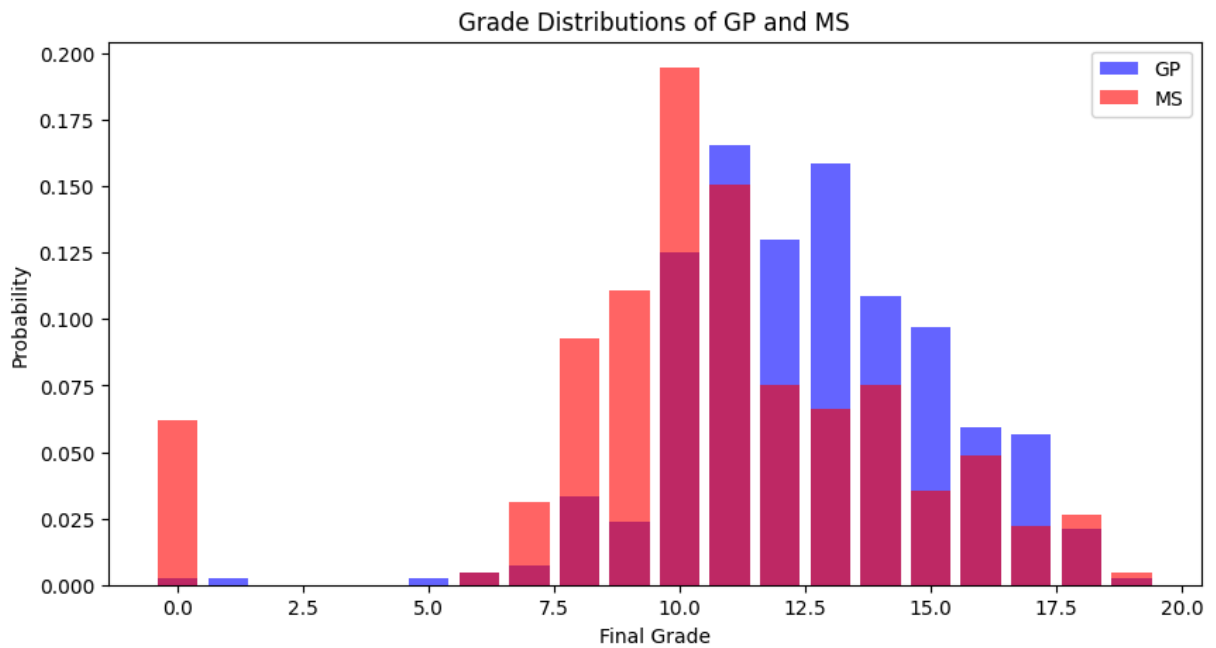
    print("(c)")
    if mean_x > mean_y:
        print("GP students performed better on average.")
    elif mean_x < mean_y:
        print("MS students performed better on average.")
    else:
        print("Both schools performed similarly on average.")

main()

```

(a)





(b)

GP: Mean = 12.58, Std Dev = 2.62

MS: Mean = 10.65, Std Dev = 3.83

(c)

GP students performed better on average.

- **Total Loss (100%):**

- Loss = \$400,000
- Probability = 0.002
- Expected Loss = (0.002 \times 400,000 = 800)

- **50% Loss:**

- Loss = 50% of 400,000 = 200,000
- Probability = 0.01
- Expected Loss = (0.01 \times 200,000 = 2,000)

- **25% Loss:**

- Loss = 25% of 400,000 = 100,000
- Probability = 0.1
- Expected Loss = (0.1 \times 100,000 = 10,000)

Now, sum up the expected losses:

Total Expected Loss = 800 + 2,000 + 10,000 = 12,800

Let P be the premium charged by the insurance company. The company wants an average profit of \$500, so the expected premium income must cover the expected loss plus the desired profit.

Thus, the equation for the premium is:

$$P = \text{Expected Loss} + \text{Desired Profit}$$

Substitute the known values:

$$P = 12,800 + 500 = 13,300$$

The insurance company should charge a premium of **\$13,300** per year to realize an average profit of \$500.